

WellComp 2025: Eighth International Workshop on Computing and Software Systems for Well-Being

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Abstract

With the advancements in ubiquitous computing, software engineering, service systems, and ubicomp technology has deeply spread into our daily lives, including office work, home and house-keeping, health management, transportation, and even urban living environments. Furthermore, beyond the initial metric of computing, such as “efficiency” and “productivity”, the benefits that people (users) benefit on a well-being perspective based on such ubiquitous technology have been greatly paid attention to in recent years. In our eighth “WellComp” (Computing for Well-being) workshop, we intensively discuss the contribution of ubiquitous computing and software engineering towards users’ well-being that covers physical, mental, and social wellness (and their combinations), from the viewpoints of various different layers of computing. After big success of the seven previous workshops, with strong international organization members in various ubicomp research domains, WellComp 2025 will bring together researchers and practitioners from the academia and industry to explore versatile topics related to well-being and ubiquitous computing.

CCS Concepts

• Human computer interaction (HCI); • Ubiquitous computing; • Software Engineering and Information Systems;

Keywords

Well-being; Physical Wellness; Mental Health; Social Good

ACM Reference Format:

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1 Background

In the advancing ubiquitous computing age, computing technology has already spread into many aspects of our daily lives, such as office work, home and house-keeping, health management, transportation, or even cities. We have been experiencing much of the influence of those technologies are both contributing to a better quality of life (QoL) of our individual and organizational lives, and causing new types of stress and pain at the same time. In the U.S., life satisfaction has not increased significantly for the last 30 years despite the gross wealth tripling, versatile computing technologies being introduced into their lives [9].

The term “well-being” has recently gained attention as a term that covers our general happiness and even more concrete good conditions in our lives, such as physical, psychological, and social wellness. Even economists and governments are starting to focus on well-being and “Gross National Happiness” as a new metric for measuring the statuses of the nations in addition to the traditional metrics such as GDP and unemployment rate. An increasing number of researchers, engineers, and people are paying attention to how their work can contribute to better quality of lives, social good, and well-being. This is quite in contrast to the criteria highlighted in the early age of computing, where the focus was on productivity and efficiency. Several trends in the technology field [1–4] are already being observed. Also, the “Positive Computing” community has initiated activities in the human computer interaction (HCI) domain [7], being inspired from positive psychology [13] that focuses on people’s psychology under healthy conditions (not in the ill-ed condition). Meanwhile, also in the UbiComp community, two workshops [5, 6] have took place, mainly focusing on detecting people’s mental illness by using smart and wearable devices.

In spite of such recent activities in academia and the society, unified academic research activities on computing, software engineering, and well-being is anticipated within the ubicomp research community. Active research not only in the HCI domain but in various other ubicomp research areas (systems, mobile/wearable sensing, mobile computing, persuasive applications and services, behavior change, etc.) are needed towards drawing the big picture of “computing for well-being” from different viewpoints and layers of computing. Researchers have been already actively working on human activity recognition (mainly focusing on “physical” activities), psychological state recognition, and even societal activity



Figure 1: Photos from WellComp 2018.



Figure 2: Photos from WellComp 2019.

recognition in urban and natural environments [8, 10–12, 15]. Additional viewpoint of “well-being” of people/community in their researches will be possible with new types of applications that comprehensively cover such different types of activity recognition. Ever since Mark Weiser [14] introduced the term of ubiquitous computing, the ubiquity of computing in our daily lives and the society has been certainly progressing. Now it is time for the community to more seriously envision the benefits that such computing technologies can bring.

2 WellComp 2025 Workshop

In this workshop, we will bring together people from industry and academia who are active in the areas of activity recognition, mental health, social good, context-awareness and ubiquitous computing. Following seven very successful workshops (WellComp 2018, 2019, 2020, 2021, 2022, 2023 and 2024) with more than 50 attendees respectively (shown in Figure 1 to Figure 4), the main objective of WellComp 2025 is to share the latest research in various areas in computing, software and system engineering, related to users’ physical, mental, and social well-being. Expected areas and topics include, but not limited to, systems research, user studies, networking, social computing, as shown in the list as follows:

- Measurement and representation of physical, mental, and social well-being with ubicomp technologies.

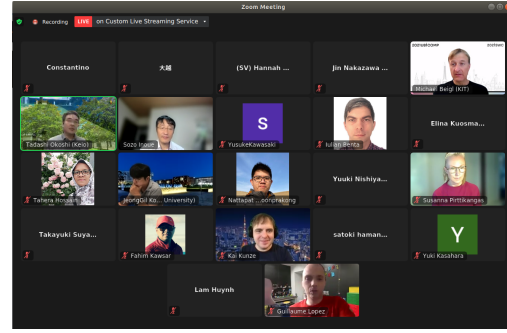


Figure 3: Screenshot from the virtual meeting at WellComp 2021 (Virtual Event).

- Design and implementation of platforms for collecting, processing, and interpreting health and well-being data.
- Design and development of computational models predictive of one or several aspects of well-being.
- Unsupervised, semi-supervised, and supervised representation learning for well-being.
- Classification, regression, and clustering problems related to well-being aspects.
- Leveraging large foundation models to improve computing for well-being.
- Exploring opportunities and challenges of LLMs in well-being applications.
- Approaches addressing challenges in wearable sensor data (e.g., missing and noisy data, irregular sampling rates, few labels, out-of-distribution inputs, etc) used for well-being.
- Development of explainable, robust, privacy-aware, and trustworthy pipelines for well-being monitoring.
- Multi-modal approaches integrating information from several data sources (e.g., physiological, behavioral, audio).
- Fairness in computing systems for wellbeing
- Ethical considerations from data collection, system development to deployment
- Computing systems for promoting well-being-awareness.
- Innovative well-being applications and diverse target populations (e.g., children, patients, or elderly people).
- Feasibility, usability, and evaluation of AI-enhanced well-being applications.
- Deployment challenges of AI software in well-being applications.

Especially for the eighth workshop, this year’s special attention will be drawn to papers addressing the “challenges for physical, social and mental well-being monitoring using ubicomp technologies and software engineering approach.” Relevance to such topics will be considered in the paper review and selection process. Furthermore, the workshop aims to identify future research challenges, research opportunities, and applications of our research outcomes to society. **Relevance and impact to UbiComp:** The emergence of versatile types of computing devices (including IoT), services, and computing venues has been realizing the advanced “ubiquity” of computing. WellComp’s focus on user well-being is related to all of the UbiComp research fields above and can open new applications to which ubicomp technology can be applied. Furthermore, the



Figure 4: Photos from WellComp 2022 in Cambridge, UK.



Figure 5: Photo from WellComp 2023 in Cancun, Mexico.

introduction of new applications and insights from the WellComp workshop also has significant potential to impact and contribute to UbiComp's interest areas horizontally.

Long-term objectives: The workshop will lead to a deeper understanding of well-being and its relationship with ubiquitous computing. The eighth workshop will position the theme within the UbiComp research community and the definition of “computing for well-being” will continuously be refined. Under such efforts, we plan to build an active and long-lasting community around the workshop's theme. For example, we want to use the workshop's drive to prepare a special issue of a journal, e.g., IEEE Pervasive Computing. We plan to make an open call for this special issue, but will especially invite workshop participants to submit their work. Furthermore, we consolidate what we find from the workshop (including discussion) into an article that highlights the workshop's main insights.

The workshop's website, along with social media accounts, will be the central venue for us to disseminate such insights from the workshop. Our website is available in <https://wellcomp2025.github.io>. We will share all workshop contributions (including papers and discussion results) on this website. Furthermore, social media pages (e.g., Facebook page) will be leveraged to reach potential authors, participants, and further collaborators.

3 Organizers

WellComp 2025 has a strong international organizing team of researchers in versatile ubicomp-related areas.

- **Hong Jia** is a Lecturer at the University of Auckland, where he is working on the intersection of Efficient ML, Mobile



Figure 6: Photos from WellComp 2024, Melbourne, Australia.

Computing and HCI. Between 2021 and 2024, he was a Research Associate in the mobile computing laboratory at the University of Cambridge. His work regularly publish in mobile computing in topics of efficient ML and mobile sensing systems for human wellbeing and pervasive computing. His research has been recognized by the ACM Mobile Computing community, and he has been awarded the ACM MobiSys Rising Star in 2023. He serves as a TPC member for WellComp 2024, a regular PC member for UbiComp, a TPC member for WWW, and a PC member for NeurIPS. He also has extensive experience in organizing and co-chairing workshops.

- **Pantea Keikhosrokiani** is a Docent and Postdoctoral Researcher at University of Oulu. She is specializing in data/AI-driven service systems and software for better decision-making. Her work integrates innovative technologies and predictive modeling to enhance user experiences, human factors, cognitive load to promote health and well-being. She is part of the 6G-enabled Sustainable Society (6GESS) to create a data-driven, hyper-connected future smart society. Furthermore, her research on optimizing user-centered design aims to align technological advancements with human needs, thereby enhancing interaction quality between users and digital platforms through intelligent systems. Pantea Keikhosrokiani is also the author of the book *Perspectives in the Development of Mobile Medical Information Systems: Life Cycle, Management, Methodological Approach, and Application* (Elsevier, 2019) and the editor of *Big Data Analytics for Healthcare* (Elsevier, 2022).
- **Soumyajit Chatterjee** is a Research Scientist at Nokia Bell Labs, Cambridge (UK). His work surrounds enabling AI on mobile and resource constrained devices to exploit the rich spatiotemporal information that these ubiquitous devices

capture. Currently, he also holds a visiting scholar position at the University of Cambridge, UK. Prior to joining Bell Labs, he was a PhD student at IIT Kharagpur. He has previously co-organized a workshop on Federated Learning for Audio Understanding (FLute) at IEEE ICASSP '25, Graduate Forum in COMSNETS '23 and was a part of the organizing committee of ICDCN '24 and COMSNETS '21. Additionally, he has served in the program committees for ACM CSCW, ACM ICMI, ACM UbiComp/ISWC, ICDCS, COMSNETS, and ACII.

- **Ting Dang** is a Senior Lecturer at the University of Melbourne, with her research primarily focusing on human-centered AI in health, acoustic modelling, time series modelling and wearable sensing. She has worked in both academia and industry and has experience understanding research from both perspectives. Additionally, she has won the Best/Top Paper awards at ACII and ICASSP. She has a rich background in organizing and co-chairing tutorials, workshops and industry perspectives at conferences of ACII, ICMI, UbiComp, and MobileHCI, and has also been successful as part of the bidding team for Interspeech 2026.

The Steering Committee of the WellComp Workshops is:

- **Tadashi Okoshi**, Associate Professor, Keio University, Japan
- **Jin Nakazawa**, Associate Professor, Keio University, Japan
- **JeongGil Ko**, Assistant Professor, Ajou University, South Korea
- **Vassilis Kostakos**, Professor, University of Melbourne, Australia
- **Fahim Kawsar**, Department Lead, Nokia Bell Labs Cambridge, UK
- **Susanna Pirttikangas**, Research Director, University of Oulu, Finland

4 Workshop Papers

Section 2 shows the possible topics of the papers we will ask the workshop authors to contribute. We will accept two types of papers *long* (6 pages) and *short* papers (4 pages) excluding the references in the double-column format. Both papers should use the ACM SIGCHI Master Article template suggested in the UbiComp/ISWC submission guidelines¹. The accepted papers will **be included in the ACM Digital Library**.

We will distribute our call for papers to various mailing lists of ubicomp-related researcher communities, including ubiquitous, pervasive and mobile computing, HCI, software engineering, systems research, IoT, distributed systems, or mobile health.

At least two reviewers from our organizers and PC members will peer-review each submission. Our acceptance criteria will be a mixture of relevance, novelty, provocativeness, and research quality. **We expect to have a selection of 8-12 accepted papers.** Given the theme of this workshop, we are confident to attract a large number of paper submissions which enables us to construct a high-quality workshop. We will request at least one author of each accepted paper to register for the conference and the workshop itself. All papers will need to be anonymized.

Table 1: The expected workshop program: a mixture of inspiring keynotes, and short and long-paper presentations.

Time	Program
09:00–09:15	Introduction
09:15–10:15	Keynote Talk 1
10:15–10:45	Paper Presentation Session
10:45–11:00	Coffee Break
11:00–11:30	Paper Presentation Session
11:30–12:30	Keynote Talk 2
12:30–13:00	Networking

5 Workshop Structure

Following seven successful workshops in 2018, 2019, 2020, 2021, 2022, 2023 and 2024 we plan WellComp'25 as a **half-day workshop** whose program is shown in Table 1. We're planning to have two keynote lectures by distinguished researchers focusing on “well-being”, inspired by the successful and interactive keynotes from WellComp 2023 and 2024. The opening keynote will be followed by two paper presentation sessions and intensive Q&A sessions. During the paper presentation sessions, each accepted paper will be presented by one of the authors. The closing keynote is followed by a closing session that summarizes the key findings from the workshop and discusses possible future collaborations and activities of the community. We will then allow the workshop participants to have a networking and interacting session at the end of the workshop. Participation in the workshop will not be limited.

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¹Guidelines: <https://www.ubicomp.org/ubicomp-iswc-2025/authors/formatting/>